

commonscore

for Invisible Garden

A formal proposal for a social-oracle, escrow-based funding primitive

Understanding the Problem

Invisible Garden, as many other pop up city residency programs, operates under tight sponsorship budgets while facing asymmetric information:

- **sponsors cannot reliably observe how funds are used on the ground** (events, residencies, hackathons), and
- **organizers lack a lightweight mechanism to demonstrate compliance** without bureaucratic overhead.

This leads to conservative budgets, delayed disbursements, and low tolerance for experimentation.

Early Web3 funding relied on informal multisigs and off-chain reporting. Programmatic mechanisms (e.g., quadratic funding, matching pools, DAO proposal systems) improved allocation but not the last-mile verification of delivery.

Initiatives like Gitcoin Grants, Optimism's RetroPGF, and POAPs helped measure participation and impact, yet there remains a gap between human confirmation of reality (what actually happened at the venue, who contributed, whether milestones were met) and automatic fund release. Attestation frameworks (e.g., EAS) emerged to bridge this gap, but are often used as metadata rather than as hard preconditions for payments.

Web3 Current Landscape

Today's tooling makes it feasible to combine:

- (i) on-chain custody (escrow),
- (ii) verifiable attestations,
- (iii) participant identity/eligibility gating, and
- (iv) privacy-aware analytics.

However, projects still stitch these pieces manually.

Escrow patterns are underused for program governance; multisigs dominate despite weak audit trails for *why* funds moved. A composable, minimal, and auditable funding primitive is still missing for live events and short-cycle programs.

Hypothesis

If sponsors can see that funds are released only when community-backed attestations confirm that commitments were met, they will increase budget allocations and risk tolerance.

In other words, the path to higher trust is to operationalize a **social oracle that turns human validation into machine-executable conditions for payments.**

Benchmark

Actors and Initiatives

The search for transparent, auditable, and automated fund management systems in Web3 has accelerated in recent years, particularly for programs such as hackathons, residencies, and pop-up cities. Below is an overview of the most relevant actors and tools currently experimenting with this space, outlining both their contributions and the remaining gaps that commonscore Escrow aims to address.

- **Bitcoin Grants / CLR (Quadratic Funding):**

A pioneering model for community-driven funding that distributes grants proportionally to the number of contributors, not the total amount donated. It effectively transforms community sentiment into funding power. However, its mechanisms stop at allocation; delivery verification and milestone validation remain off-chain, relying on human reporting rather than automatic condition checks.

- Optimism RetroPGF (Retroactive Public Goods Funding):**
 Distributes funds based on post-factum attestations of impact by trusted reviewers. It is one of the most credible “social validation” experiments at scale.
- Arbitrum Grant Program Allocator:**
 Requires milestone definition (1–5 per grant), but verification is admin-led and centralized. There is no trustless or programmable validation layer between milestone review and fund release.
- DoraHacks Ecosystem:**
 One of the leading infrastructures for hosting Web3 hackathons and bounties. DoraHacks implements Quadratic Funding, DAO-based voting, and privacy-preserving voting tools (MACI). Still, these tools focus mainly on pre-award allocation and post-event prize release, rather than continuous conditional escrow during execution.
- Gnosis Safe / Safe Global:**
 The dominant multisig and treasury management standard in Web3. Enables shared custody of funds and collective approval for transactions. However, signatures represent consent—not evidence of real-world delivery. Validation remains administrative, not programmatic.
- Ethereum Attestation Service (EAS):**
 Provides a unified schema for attestations on-chain. While widely adopted for identity, trust, and credential systems, EAS is typically used for data recording, not as an enforcement layer for conditional fund release. Its potential as a social oracle remains largely unexplored.
- POAP (Proof of Attendance Protocol):**
 Used for attendance verification and participation tracking. While POAPs provide proof of presence, they do not establish proof of performance or task completion—critical for accountability in funding programs.
- Soulbound Tokens (SBTs):**
 Non-transferable tokens that represent achievements, reputation, or membership. They enable eligibility-based payments (e.g., only hackathon winners receive airdrops), yet lack flexibility for dynamic, conditional fund flows that depend on multiple actors’ validation.

- Coordinape / Karma / CharmVerse:**
 Platforms for peer-based contribution tracking and reputation-weighted rewards. These systems provide structured “peer attestations,” aligning well with the logic of a social oracle, but they are application-layer tools, not protocol-level escrows. Payments are executed manually or through DAO votes.
- RetroPGF and MetaGov ecosystems:**
 Emerging frameworks for retroactive evaluation and governance-based accountability. These systems rely on curated reviewer networks rather than distributed attestations by participants themselves.
- Circles (Social Trust Network):**
 Introduces personal currencies and trust graphs to model human-based credit and authenticity. Although not built for escrow logic, Circles demonstrates how trust relationships can serve as programmatic conditions—an approach that commonscore Escrow extends through its Social Oracle model.

Category	Tool / Protocol	Function	Limitation
Fund Allocation	Bitcoin CLR, DoraHacks QF	Community-driven allocation based on votes	No delivery verification
Retroactive Evaluation	Optimism RetroPGF, MetaGov	Post-outcome funding based on attestations	No milestone-level execution
Escrow and Treasury	Gnosis Safe	Secure custody and shared approvals	Manual, not condition-based

Attestation Infrastructure	EAS, Verax, Attest.sh	On-chain attestations and schemas	Data layer only, not fund-controlling
Identity and Presence	POAP, SBTs, Lens Profiles	Proof of participation or role	Do not prove fulfillment
Reputation and Peer Validation	Coordinape, Karma, SourceCred	Peer-based scoring and reward allocation	Off-chain, manual payouts
Social Trust Graphs	Circles, BrightID	Human verification and trust edges	Limited liquidity, non-transactional validation
Privacy and Fairness	MAPI, Snapshot X	Anti-collusion and encrypted voting	Focused on governance, not fund release

While the ecosystem has evolved a rich set of tools for fund allocation, identity, and governance, few have bridged the gap between human validation and automated fund execution.

Most rely on social signals for *deciding who should receive funds*, but not for *confirming that obligations were fulfilled*.

Our proposed commonscore escrow addresses this gap by combining three proven primitives (escrow contracts, attestations, and social trust graphs) into a single mechanism that allows money to move only when people and code agree that reality occurred.

Field Research

Insights from Sponsors, Community Managers, and Hub Coordinators

To understand whether a conditional-funding mechanism like commonscore Escrow would be useful in real-world contexts, we conducted a series of semi-private conversations with individuals involved in *Avalanche Latam*, *multi-community coordination in Latin America*, *Invisible Garden*, *Arbitrum's grants process*, *permanent hubs such as Crecimiento*, *The Mu*, *Zuzalu*, and *Ethereum Everywhere*. These interviews were informal and exploratory; because they were not collected under formal research protocols, we avoid quoting or attributing statements directly. Instead, we group the insights into themes that appeared consistently across discussions.

Lack of Clarity and Transparency in Event Funding

Across multiple sponsors and organizers, one recurring theme was that estimating budgets is extremely difficult, especially for events taking place in regions where sponsors do not have direct cost familiarity. This creates friction when evaluating whether the proposed expenses are realistic or inflated. Most sponsors expressed that they rarely have visibility into how funds are ultimately used, and while some do not require granular oversight, others said that more transparency could improve their confidence and willingness to fund recurrently.

The Challenge of Upfront Costs

A key operational constraint mentioned by nearly every event organizer is that suppliers often require upfront payment, especially for catering, venue rental, equipment, or cleaning services. Any mechanism that restricts all payments until after validation would be unworkable. Instead, organizers highlighted the need for hybrid models: partial upfront payments to unlock execution capacity, followed by conditional releases tied to evidence of fulfillment.

Difficulty Accessing Sufficient and Consistent Funding

Both pop-up city coordinators and permanent hub operators reported the same structural problem: fundraising is unstable, and current models make it hard to cover essential operating costs or compensate staff. Even projects with strong community value struggle because sponsors prefer hosting their own branded events instead of contributing to shared infrastructures. Several organizers voiced that collective efforts tend to be underfunded simply because recognition is harder to attribute.

Permanent hubs added an extra concern: beneficiaries often value “free access” but lack mechanisms to share responsibility for sustaining the space financially.

Need for Better Milestone Verification and Accountability

Grant-related roles, particularly from ecosystems like Arbitrum, emphasized that transparency and accountability are important but difficult to enforce with current tools. Milestones are often generic, administrative, or reviewed by internal teams without input from people on the ground. There is a strong desire for more ground-truth verification that includes the perspective of participants or attendees, not just organizers or administrators.

Importance of Mission Alignment and Local Impact

Pop-up city organizers from initiatives like The Mu highlighted that successful fundraising depends not only on budgets or deliverables but on clear mission alignment. Sponsors are more willing to participate when the purpose is explicit, the intended impact is concrete, and the event integrates local stakeholders (e.g., local providers, community partners) to avoid issues like gentrification. Transparent tools help reinforce this alignment by showing how resources flow into the local ecosystem.

Fragmentation, Politics, and Lack of Shared Standards

A recurring issue, especially among global pop-up city networks, is that there is no common standard for how hubs should operate, validate outcomes, or manage funds. This creates internal politics, competition for legitimacy, and inconsistent practices. Some communities are included in “official directories,” while others—equally active—are left out without a clear rationale. This reinforces the need for neutral, credible, and open frameworks that reduce discretionary gatekeeping.

Value of Including Participants in Validation

Several organizers emphasized that people on the ground often have the most accurate perspective on whether something was delivered as promised. Whether evaluating a hackathon milestone, the quality of a meal, or the effectiveness of logistics, participant input offers a realistic indicator of fulfillment that complements organizer reporting. This reinforces the usefulness of attestation-based validation systems, especially when they can aggregate multiple human signals into a unified, auditable conclusion.

Across all interviews, three broad needs emerged clearly:

1. **More transparent and credible mechanisms to show sponsors how funds are being used.**
2. **Hybrid payment structures that support both upfront liquidity and conditional verification.**
3. **Distributed validation processes that include organizers, participants, and sometimes local stakeholders, rather than relying solely on centralized approval.**

These patterns strongly validate the core premise of the commonscore escrow model: that fund release should depend on verifiable social confirmation across multiple roles, not on trust in a single entity.

What Is Missing

- **Conditioned escrow primitive:** payments blocked until attestations reach a quorum defined ex-ante.
- **Hierarchical agreements:** parent program with sub-agreements per provider/track, each with tailored release logic.
- **Participant-as-attestor flows:** eligibility via NFT + group membership, fast mobile attestation during/after the event.
- **Retro funding sink (RetroVault):** default destination for failed/unused conditional funds, enabling retrospective rewards.
- **Composability:** clean interfaces for POAP, EAS, Circles groups, and optional reputation modules without lock-in.

Our Proposal

A) Social Oracle

We refer to "social oracle" as the aggregated set of human attestations that confirm whether a commitment was honored (e.g., catering delivered on time and quantity; venue open and functional; participant milestone achieved).

This signal becomes a hard precondition for fund release. The oracle is not a subjective narrative; it is a structured, signer-scoped, and schema-bound data flow that the contract verifies.

Design implications:

- Attestation schema per agreement (**agreementId**, **attestor**, **decision**, **timestamp**, **proofOfEligibility**).
- Eligibility checks: possession of the program NFT, membership in a Circles Group, and optional POAP check-in.
- Quorum function configurable per sub-agreement (e.g., $\geq 60\%$ of eligible attestors or $\geq N$ confirmations).

The **Social Oracle** converts collective human perception into programmable truth.

- Each sub-agreement defines validation moments (e.g., meal completion, milestone delivery).
- When these occur, an **on-chain event** triggers notifications to a randomized subset of eligible wallets—those holding the participant NFT, Circles Group membership, and a POAP proving presence.
- Notifications are sent through open-source tools like **Push Protocol**, **XMTP**, or **Lens Messaging**; participants respond through a lightweight attestation interface (React/Next.js) signing a binary prompt such as *"Was the catering delivered on time and as agreed?"*

Eligibility Criteria:

To prevent spam and ensure contextual relevance, a wallet must satisfy:

- Ownership of a Participant NFT (proves acceptance into the program).
- Membership in the event's Circles Group (socially verified identity).
- POAP attendance proof (indicates physical or digital presence at the event).

Randomized Selection:

From the eligible pool, a pseudo-random subset (e.g., 10–20%) is selected for each validation round using a deterministic algorithm:

```
uint256 seed = uint256(keccak256(abi.encodePacked(block.timestamp,  
agreementId)));
```

```
selectedIndex = seed % eligibleCount;
```

This randomness, based on public parameters, ensures that attestors cannot be pre-targeted or bribed.

Anonymity:

- Attestations store only hashed wallet identifiers; individual votes remain private.
- Aggregated quorum percentages are public and auditable.
- Optional zk-attestations (via EAS ZK SDK) can provide zero-knowledge proof of valid participation without revealing the wallet address or decision.

Attestations are recorded via **EAS**, containing minimal fields: agreement ID, prompt ID, decision, timestamp, and hashed proof of eligibility.

- Attestors are pseudo-randomly selected; their inputs remain pseudonymous or private (zk-attestations optional).
- When the quorum threshold is reached, the **EscrowAgreement** releases funds; if unmet before the deadline, funds redirect automatically to the **RetroVault** for retroactive redistribution.
- No payment occurs without verifiable human confirmation that commitments were fulfilled.

Each sub-agreement reserves 1–2% of its budget as a **validation bounty**, paid equally to all valid attestations (regardless of outcome).

- Rewards come from the **RetroVault**, incentivizing participation rather than bias.
- Validators accumulate a **Trust Score** within Circles through “validation edges” in the trust graph, strengthening their weight in future rounds.
- Circles thus evolves from a static membership layer into a **dynamic reputation network** for program reliability.

Collusion resistance emerges from random validator selection, eligibility gating (NFT + Circles + POAP), anonymity, and reputation weighting.

- Manipulation would require controlling multiple verified wallets—economically infeasible in practice.
- Sponsors or hubs can audit attestations, trigger meta-validation rounds, and analyze variance to detect anomalies.

Compared with prior models (multisigs, manual organizer approvals), the commonscore **Social Oracle**:

- Enforces funding conditions directly on-chain.
- Rewards honest participation with neutral micro-bounties.
- Captures human confirmation as verifiable state data.
- Integrates **identity, perception, validation, and execution** into a single modular primitive where *funds move only when people—and code—agree that reality occurred*.

B) Factory Smart Contract

A minimal factory contract deploys parameterized EscrowAgreement instances.

Factory responsibilities:

- Register the root program (hub, sponsor, budget, deadlines, oracle parameters).
- Deploy child EscrowAgreement contracts with their own payment schedule and quorum rules.
- Emit events for indexers/dashboards.

EscrowAgreement responsibilities:

- Hold funds (ETH or ERC-20) and track state transitions: **Locked** → **Fulfilled** → **Released** or **Locked** → **Failed** → **RetroVault**.
- Accept attestations, check quorum/eligibility, and execute **release()** or **routeToRetroVault()**.
- Support mixed payment terms: **upfront**, **conditional**, **mixed** (e.g., 70/30 catering model).

C) Retro Funding (RetroVault)

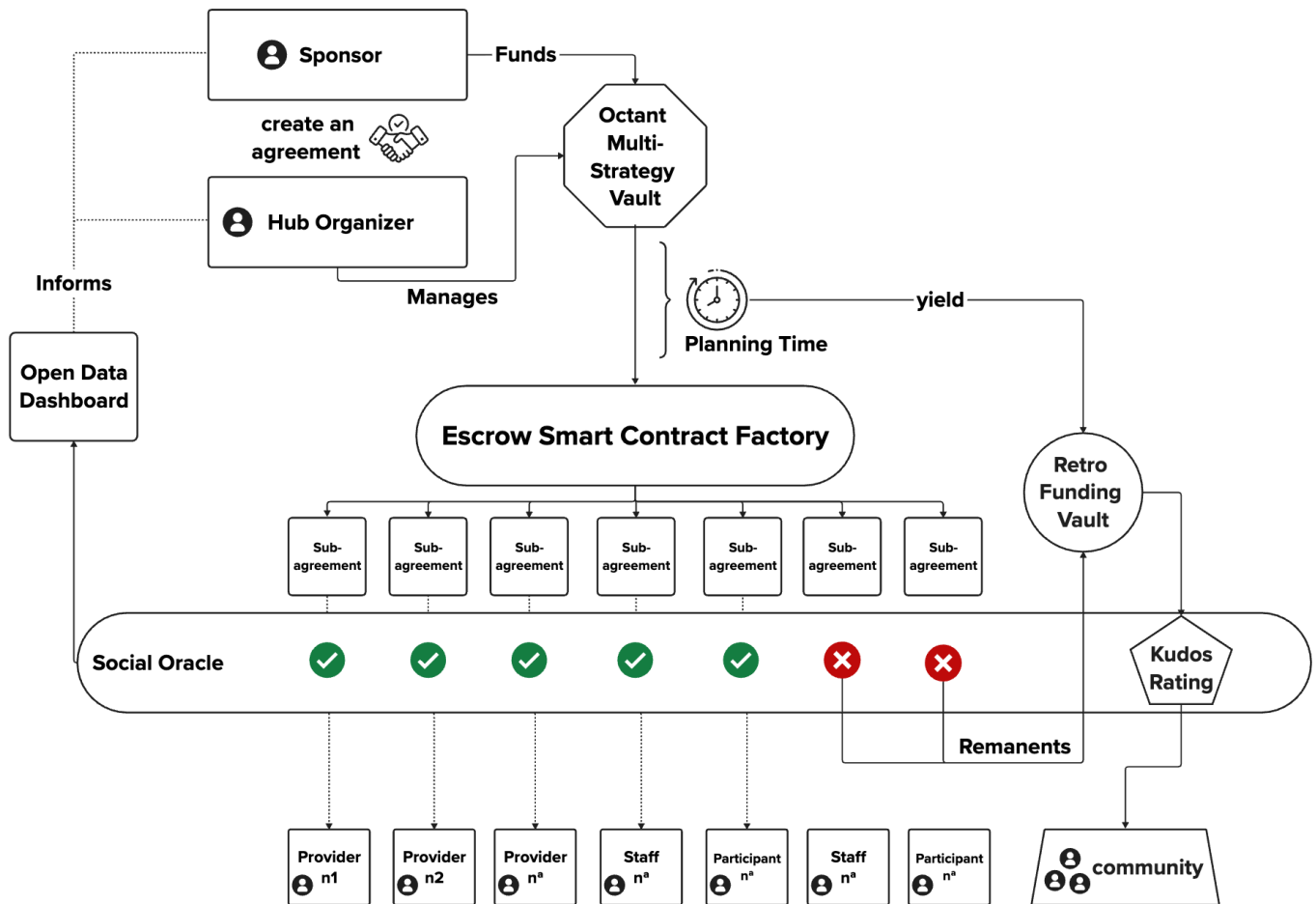
A program-scoped RetroVault serves as the sink for conditional funds that were not released (missed quorum, deadline, or cancellation). Sponsors may also pre-allocate a percentage directly to the RetroVault at program start.

Use:

Later, the hub and/or sponsors run a retrospective round to reward unplanned but valuable contributions (staff overtime, participant support, emergent needs). Releases from the RetroVault also require attestations, ensuring consistency with the social-oracle principle.

Tech Stack (Minimal, Modular)

- **Smart contracts:** factory + EscrowAgreement + RetroVault; interfaces for eligibility checks.
- **Attestations:** EAS schemas for AgreementCreated, MilestoneValidated, RefundOrRetro.
- **Identity & gating:** program NFT; Circles Group membership; optional path-based trust thresholds.
- **Attendance:** POAP or equivalent check-in.
- **Frontend:** Next.js + wallet connectors; mobile attestation flow; sponsor dashboard.
- **Backend (optional):** Supabase for indexing, caching, notifications; no business logic critical path off-chain.
- **Analytics:** event logs + optional zero-knowledge aggregation for privacy-preserving metrics.



Use Cases

Hackathons

- **Structure:** parent agreement (event) with sub-agreements for venue, catering, prizes, and cleaning.
- **Flow:** sponsors deposit → participants receive NFT → POAP at entry → open attestation after each block → quorum unlocks conditional tranches. Unused conditional funds auto-route to RetroVault.
- **Value:** granular accountability; prize disbursements conditioned on organizer attestations; vendor fairness with mixed terms (e.g., 70% upfront, 30% based on participant satisfaction).

Events

- Structure: simplified parent with 2–3 sub-agreements (venue, F&B, production).
- Flow: rapid mobile attestations by attendees; small conditional tails motivate service quality; fallback to RetroVault.
- Value: sponsors see exactly what was fulfilled; organizers optimize service levels without over-collateralizing budgets.

Pop-Up Cities

- Structure: multi-week program with rotating tracks; sub-agreements per track and per critical provider.
- Flow: periodic attestations by cohort participants; conditional tails ensure continuity of quality; RetroVault captures unspent conditional funds for retroactive community support.
- Value: sustained transparency for long-horizon sponsorships; comparable metrics across tracks.

Permanent Hubs

- Structure: recurring agreements for operations with smaller conditional tails and a larger RetroVault allocation.
- Flow: monthly/quarterly attestations; RetroVault funds retroactive recognition cycles.
- Value: predictable operations plus ongoing accountability and community-aligned retro rewards.

Last words

commonscore provides Invisible Garden with a minimal, auditable, and modular funding primitive that connects human validation to automated disbursements.

By centering the social oracle, using a factory-deployed escrow architecture, and reserving conditional tails plus a RetroVault, the program achieves higher sponsor confidence, fair vendor terms, and credible, reusable infrastructure for future initiatives.

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